

TOOBA LIAQUAT

MS Computer Science | Research Assistant | AI Engineer

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PROFESSIONAL SUMMARY

Data Scientist and MS Computer Science graduate (COMSATS University Islamabad) with over two years of applied research and industry experience in machine learning, deep learning, and data analytics. Research Assistant experience includes co-authoring two manuscripts (currently under peer review) on deep learning-based image segmentation. Combines a strong academic research foundation with hands-on teaching-adjacent skills in mentoring junior analysts, technical writing, and clear communication of complex concepts to technical and non-technical audiences. Seeking a Lecturer position in Computer Science to contribute to curriculum delivery, student mentorship, and applied AI/ML research at COMSATS University Islamabad.

EDUCATION

MS in Computer Science

Sep 2024 – Jun 2026

COMSATS University Islamabad — Islamabad, Pakistan

BS in Computer Science

Feb 2020 – Feb 2024

COMSATS University Islamabad — Islamabad, Pakistan

RESEARCH EXPERIENCE

Research Assistant

Sep 2024 – Jun 2026

COMSATS University Islamabad, Attock Campus | Supervisor: Dr. Mariam Akbar, Associate Professor, Dept. of Computer Science

- Developed the research project “A Knowledge Distillation-Based Lightweight Framework for Open-Set Instance Segmentation of Small and Occluded Weeds in Cotton Fields,” designing and evaluating deep learning models for object detection and instance segmentation.
- Co-developed the research project “Enhancing Image Segmentation Accuracy in Complex Underwater Environments,” applying deep learning and computer vision techniques to challenging real-world imagery.
- Conducted literature reviews, experimental design, and implementation of deep learning algorithms for object detection and segmentation tasks.
- Performed data collection, preprocessing, and analysis for ML experiments, generating insights that informed model design decisions.

PUBLICATIONS (UNDER REVIEW)

- “A Knowledge Distillation-Based Lightweight Framework for Open-Set Instance Segmentation of Small and Occluded Weeds in Cotton Fields.” Manuscript under review.
- “Enhancing Image Segmentation Accuracy in Complex Underwater Environments.” Manuscript under review.

PROFESSIONAL EXPERIENCE

Data Scientist / ML Engineer — Self-Employed (Freelance)

Mar 2024 – Present

Islamabad, Pakistan

- Developed, validated, and deployed predictive machine learning models (Random Forest, XGBoost, Neural Networks) to solve real-world problems, improving predictive accuracy and model reliability.
- Built end-to-end data pipelines and automated analytical workflows, continuously improving methodologies and reducing manual processing time.
- Delivered data visualizations and reports using Python, Power BI, and Tableau tailored for technical and non-technical audiences.
- Applied NLP techniques for text data analysis, feature engineering, and insight extraction across multiple projects.

INTERNSHIP

App Development Intern — CodSoft (Virtual Internship)

Sep 2023 – Dec 2023

- Completed a 12-week virtual internship in App Development, delivering assigned tasks and projects with strong performance remarks.

CORE COMPETENCIES

Academic Research & Literature Review • Technical & Manuscript Writing • Curriculum-Relevant Subject Matter (AI/ML, Data Science) • Mentoring & Team Collaboration • Predictive Model Development & Validation • Data Collection, Cleaning & Preprocessing • Exploratory Data Analysis (EDA) • Technical & Non-Technical Communication

TECHNICAL SKILLS

Programming Languages: Python, R, SQL

Machine Learning & AI: TensorFlow, Deep Learning, Computer Vision, Natural Language Processing (NLP), Random Forest, XGBoost, Neural Networks

Data Visualization & Analytics: Power BI, Tableau, MS Excel, SPSS

Cloud & DevOps: AWS, Docker, Git

Database Management: SQL, PostgreSQL

LANGUAGES

English (Professional) • Urdu (Native)